文献汇报

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Deep Learning for Generic Object Detection: A Survey

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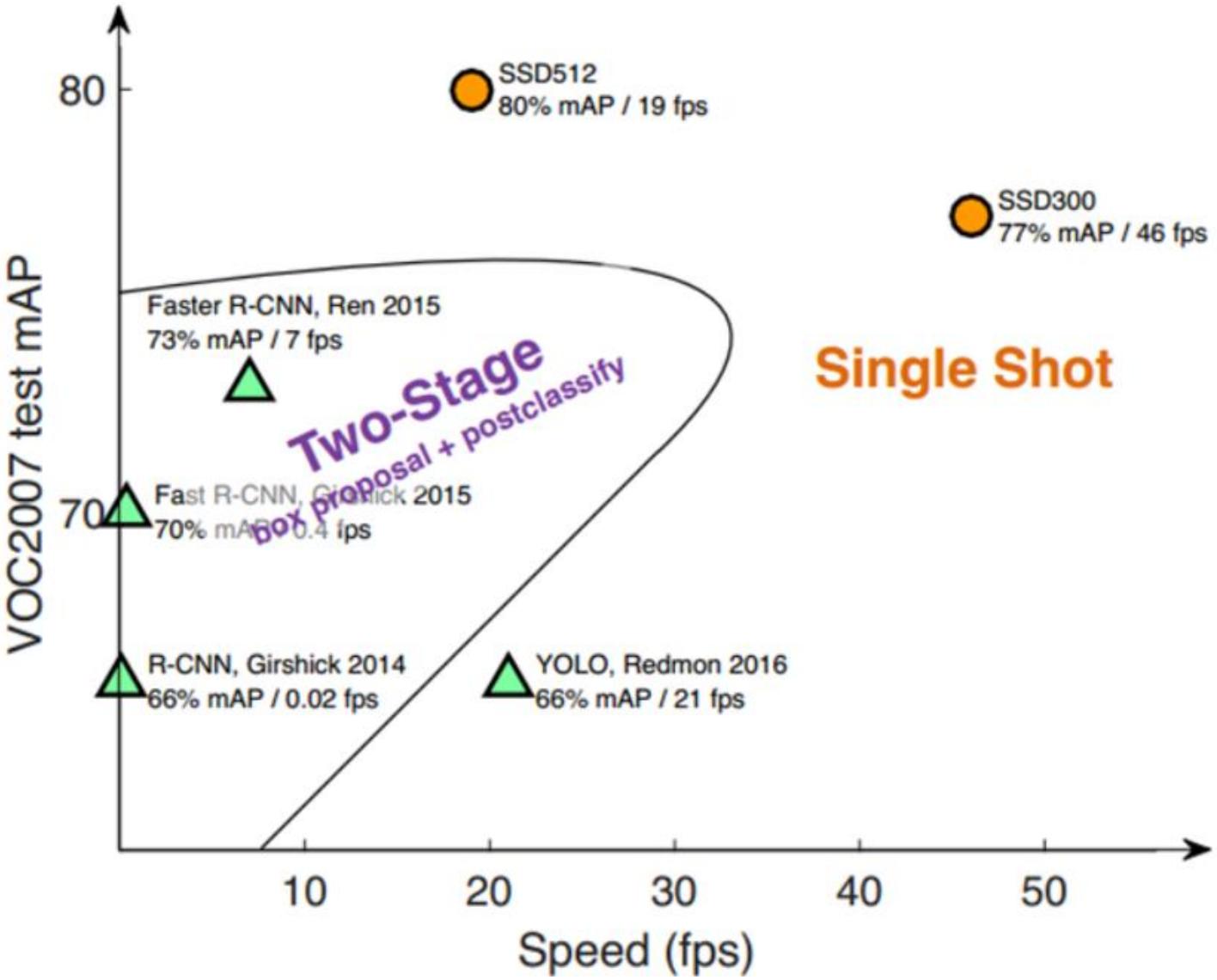
https://github.com/hoya012/deep_learning_object_detection#2014


SSD512
80% mAP / 19 fps
SSD300
77% mAP / 46 fps
Faster R-CNN, Ren 2015
73% mAP / 7 fps
Two-Stage
box proposal + postclassify
Single Shot
Fast R-CNN, Girshick 2015
70% mAP / 0.4 fps
R-CNN, Girshick 2014
66% mAP / 0.02 fps
YOLO, Redmon 2016
66% mAP / 21 fps
VOC2007 test mAP
80
70
10
20
30
40
50
Speed (fps)

一阶段：基于单次目标

一阶段的网络，整个生成备选框的过程一体化的实现。代表有

YOLO系列和SSD（single shot detector）

DetectorNet

OverFeat

YOLO

YOLOv2 and YOLO9000

SSD(Single Shot Detector)

OverFeat

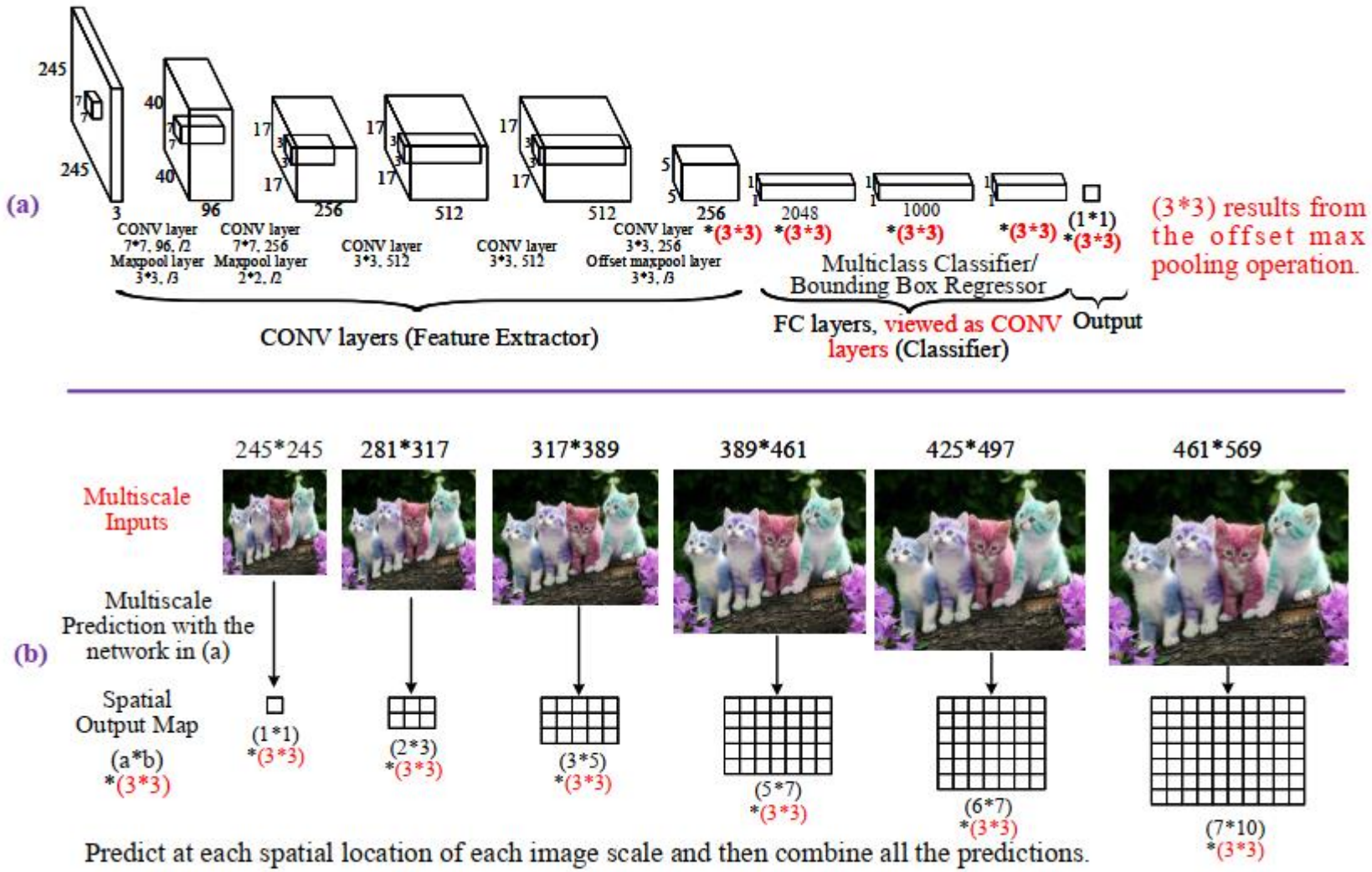


Fig. 14 Illustration of the OverFeat [239] detection framework.

OverFeat

特点：

1.滑动窗口方式获得候选对象

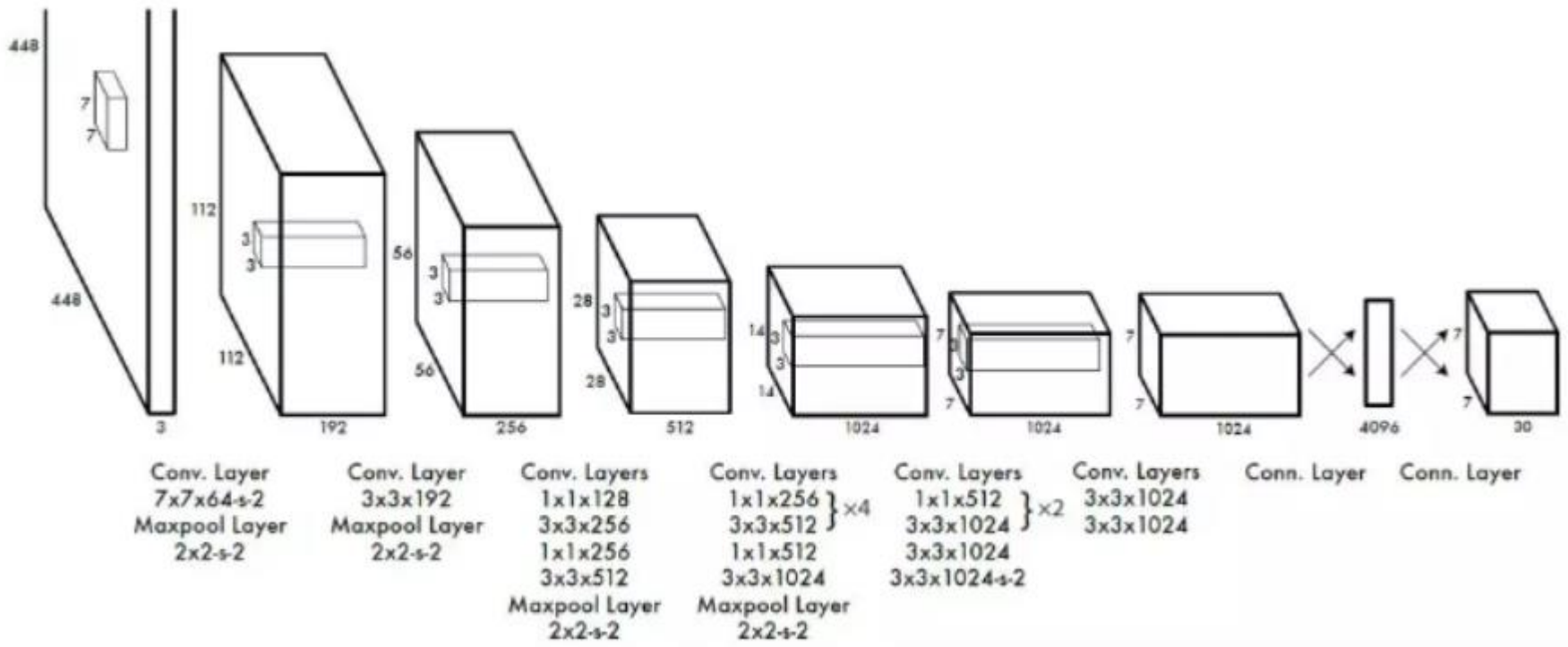
2.将FC层视为卷积层

3.通过最大偏移池化增加预测数量

4.边界框回归

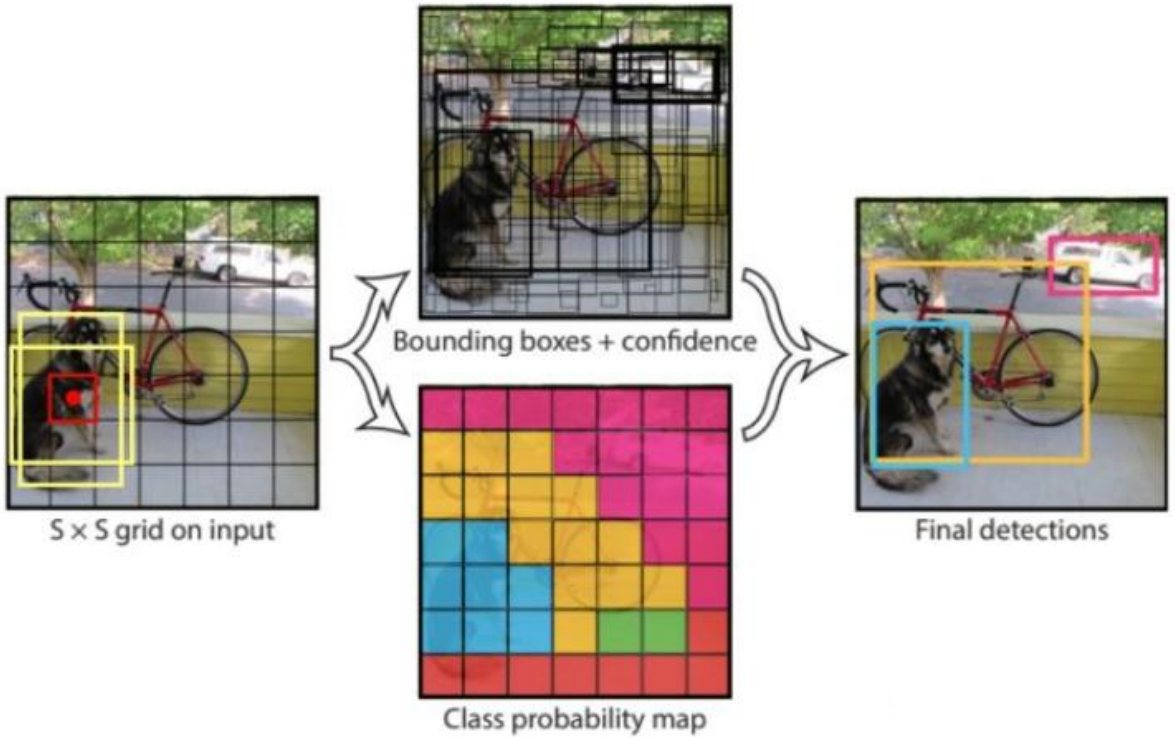
有速度优势，比RCNN准确率低

YOLO



1.没有显示地求取候选区域的模块
2.训练和检测都在一个网络中进行

YOLO



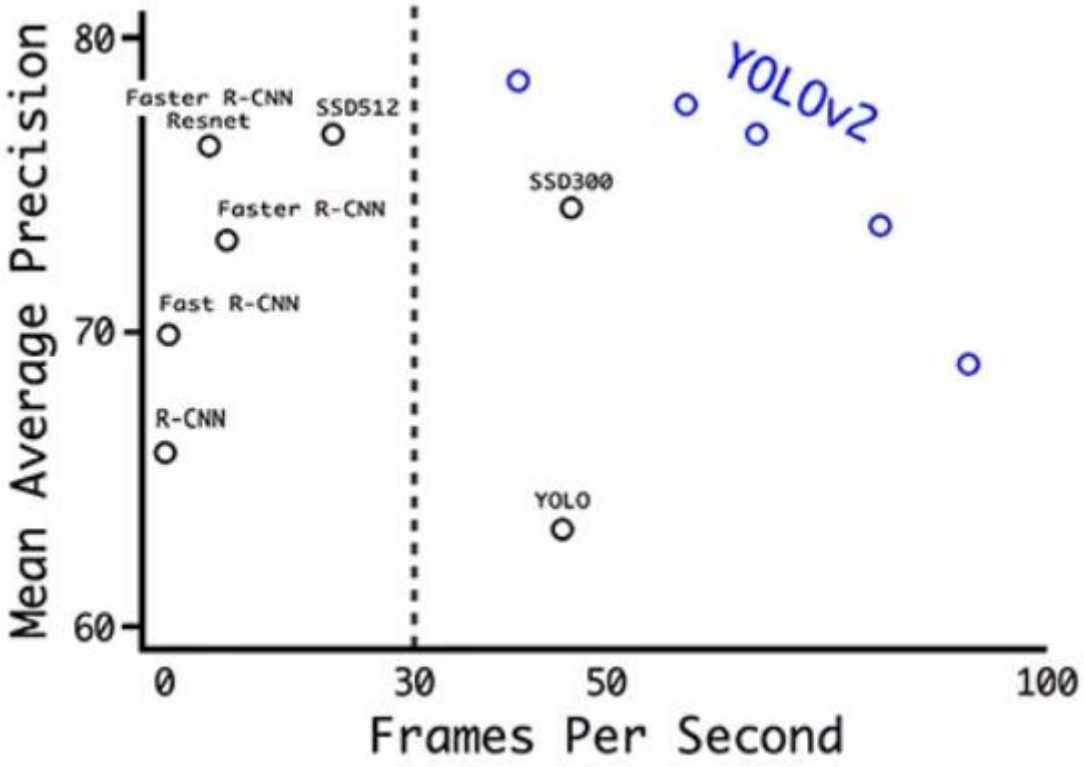
思想：
S*S小格，物体中心点落在格子里，该格子用来检测物体。
每个格子给出B 个边框，并给出边框的置信度

YOLO V2

	YOLO								YOLOv2
batch norm?		✓	✓	✓	✓	✓	✓	✓	✓
hi-res classifier?			✓	✓	✓	✓	✓	✓	✓
convolutional?				✓	✓	✓	✓	✓	✓
anchor boxes?				✓	✓				
new network?					✓	✓	✓	✓	✓
dimension priors?						✓	✓	✓	✓
location prediction?						✓	✓	✓	✓
passthrough?							✓	✓	✓
multi-scale?								✓	✓
hi-res detector?									✓
VOC2007 mAP	63.4	65.8	69.5	69.2	69.6	74.4	75.4	76.8	**78.6**

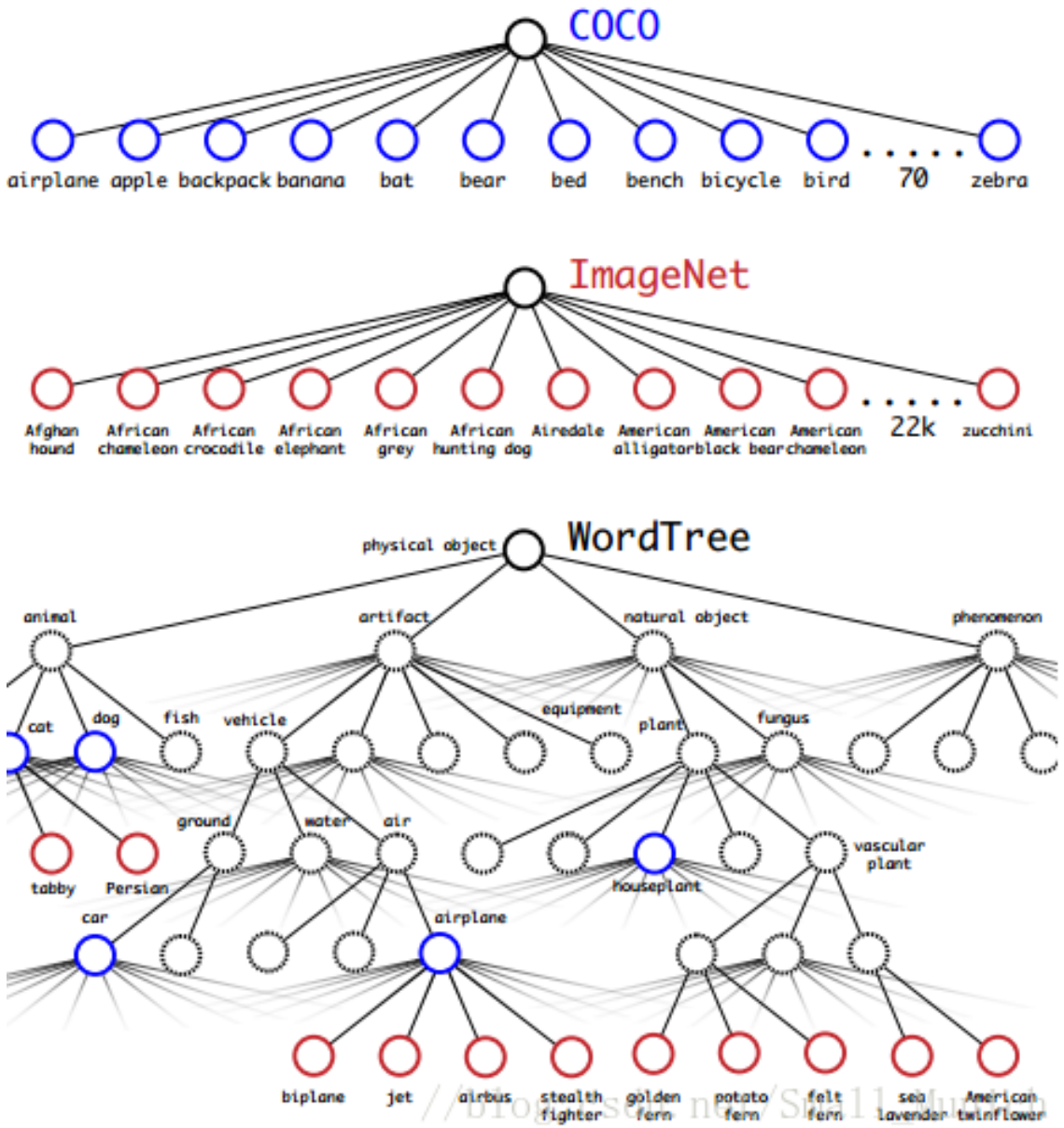
1.batch normalization（批归一化）
2.使用高分辨率图像微调分类模型
3.采用先验框（Anchor Boxes）
4.聚类提取先验框尺度
5.多尺度图像训练

YOLO V2



VOC2007数据集上目标检测算法速度与准确度对比

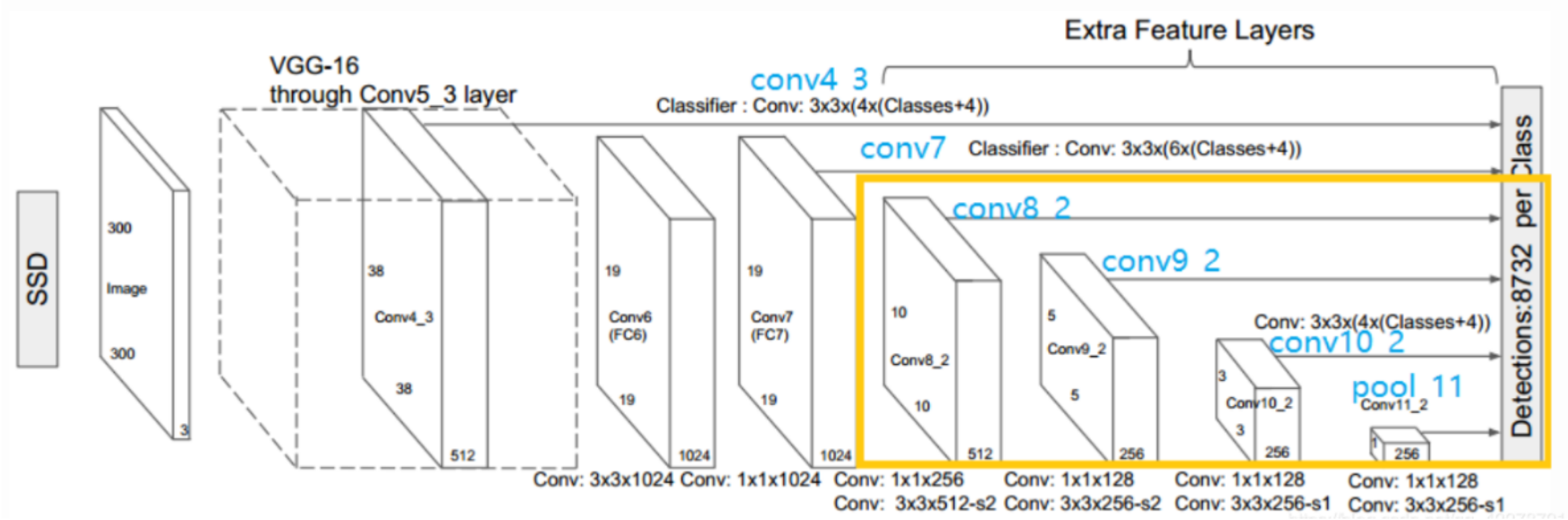
YOLO 9000



1. WordTree 进行训练

2. 使用Anchor boxes取代原始网格化的方法，使用K-Means聚类来选择Anchor boxes的大小，而不是手动选择采集

SSD（single shot detector）



借鉴VGG16，改变了第5个block的第4层，之后加入卷积层

1. RPN思想:Defalut box,后续通过softmax分类+bounding box regression获得真实目标的位置

3. NMS(非极大值抑制):可以优化多个feature map 最后会产生大量的错误BB

2. Regression思想

Thank You!